

THE OPPORTUNITY IN ONE LINE

What RLI wants: get submissions and policies in the door, read them, structure the data, decide which to work first, route to the right underwriter, and auto-adjudicate the simple ones. This is a named, well-mapped pattern in commercial underwriting: **submission ingestion and triage**, inside the broader submit-to-quote workflow.

WHY BUILD IN-HOUSE

Generic platforms work "okay" then need heavy modification for surety and specialty lines. Owning the system fits RLI's unique book better than any catch-all.

WHY NOW

Agentic tooling has lowered the build barrier dramatically. The calculus has shifted from "buy a vendor" toward "build our own, with the right partner."

WHERE STELLO FITS

An embedded, AI-native build team: US engineers who deploy custom systems on RLI's stack, not a consulting firm learning AI on the job.

TALK TRACK · DISCOVERY QUESTIONS TO ASK THEIR IT TEAM

- **Intake:** Is submission intake still email-first today, or do brokers use portals / APIs / SFTP? What volume per month, per line?
- **Systems of record:** Where does submission and policy data live now, and how many separate systems does an underwriter touch to clear one risk?
- **Legacy surface:** Which cores are green-screen / mainframe (you referenced "WINS"), and which of them expose an API versus needing screen-level automation?
- **Surety data:** Where do Work-in-Progress schedules and contractor financials land today, and how are they keyed in and reconciled?
- **Current automation:** What is your straight-through / auto-adjudication rate now, and what have you already automated that we should build around, not replace?
- **Guardrails:** What are your data-residency, security, and model-governance requirements for anything AI-driven in underwriting?

HOW WE POSITION · THE FOUR DIFFERENTIATORS

- **The data plane is the moat.** Point solutions are easy; the hard, durable value is unifying siloed systems-of-record into one queryable model. Once that layer exists, every future initiative (multi-department, cross-line) builds on top of it far faster.
- **US-based engineering, two dedicated per deployment.** Bay-Area engineers who have shipped in production environments, embedded weekly with your team. Not offshore-arbitraged talent, and not a big-firm bench.
- **You own everything.** Data, prompts, decision logic, and knowledge bases stay in-house and belong to RLI. Nothing is routed to a foundation-model vendor to train on. Built to your security bar.
- **Custom for surety and specialty.** The schema and rules are built for how RLI actually underwrites, which is exactly where a one-size platform underperforms.

VOCABULARY · SO IT READS NATIVE TO UNDERWRITERS

Submission broker request to quote a risk (email + attachments).

Clearance / dedup is this new, a duplicate, or tied to an existing deal; plus broker-authority check.

ACORD forms the standard commercial application forms and data standard.

Loss runs the insured's historical claims report.

SOV Statement of Values: schedule of insured assets and values.

Appetite the set of risks the carrier wants; triage scores fit against it.

Declination a decline-to-quote (modern systems auto-decline out-of-appetite).

Bind agreeing to provide coverage; the deal closes.

STP straight-through processing: full automation, no human touch.

Underwriting workbench the underwriter's cockpit that consolidates submissions for clearance and workflow.

Submit-to-quote the end-to-end flow, split into submit-to-triage and triage-to-quote.

Reference · the technical backbone

What to be ready to whiteboard if their IT team goes deep

VERIFIED

25 sources · 24/25 claims
adversarially confirmed

REFERENCE ARCHITECTURE · EIGHT SWAPPABLE LAYERS

Intake. Multi-channel by design: monitored mailbox (email is dominant) plus SFTP, API, upload UI. Attachments burst to object storage and a queue.

Classification. Label each attachment (ACORD 125 vs loss run vs SOV vs financials vs WIP) using few-shot LLM prompting.

IDP / OCR. Read pixels/PDF into text, tables, key-value pairs, handwriting (e.g. AWS Textract). Clean digital PDFs can go straight to a multimodal LLM.

LLM extraction. Map read text into a typed JSON schema via tool-use / structured output, with per-field confidence.

Normalization / data plane (the moat). Collapse heterogeneous inputs into one canonical "golden schema" per line. This is the layer a generic vendor cannot tailor to surety for you.

Clearance / dedup. Entity resolution against the existing book (named insured, FEIN, address, broker) to decide new vs duplicate vs renewal.

Appetite / scoring. Encode RLI guidelines as rules + a model. Phased rollout: capture 5-7 key fields to filter at intake, expand to 10-20 with enrichment, then full risk profiling.

Routing, HITL, STP, write-back. Auto-decline clear misses, STP the clean simple risks, route the rest to a confidence-ranked queue, then write structured output back to legacy cores (API where possible, RPA / 3270 screen automation where not).

SURETY & SPECIALTY · WHY IN-HOUSE IS JUSTIFIED

- **Three-party structure:** the surety guarantees the obligee that the principal will perform. Two categories: contract and commercial. Principal, obligee, bond type, and bond amount are first-class entities that do not exist in a P&C schema.
- **The WIP schedule is the crown jewel.** The Work-in-Progress schedule (per-project contract amounts, estimated costs, progress-to-date, under- and over-billings) is the surety equivalent of loss runs. Dynamic and project-level, not a static balance sheet. Often messy contractor Excel/PDF, a high-value extraction target.
- **Bonding capacity** (single and aggregate) is computed from the WIP plus financials. Triage must calculate remaining capacity, not just classify.
- **Trucking / transportation** needs a live FMCSA CSA/SMS data feed. 92% of commercial trucking underwriters use it in pricing, a different enrichment source than property or casualty.

ACCURACY & COMPLIANCE · TABLE STAKES FOR A PUBLIC CARRIER

- **Extraction is non-deterministic.** Never treat LLM output as exact. Wrap every extraction in schema validation, type/range checks, per-field confidence, and retries.
- **Confidence-thresholded human-in-the-loop.** Auto-process only above a confidence bar (higher for surety financials than for a name field); route the rest to an underwriter.
- **Audit trails.** Log inputs, prompt, model version, extracted fields, confidence, and any human override, enough to reconstruct any AI-influenced decision. Chain-of-thought reasoning kept for explainability.
- **NAIC model bulletin** on "Use of AI Systems by Insurers" is adopted in 24 states and is effectively the national standard. Governance, testing, documentation, and bias controls belong in the design from day one.

SUGGESTED FIRST BUILD · "WE'RE NOT STARTING FROM ZERO"

A **surety WIP + financials extractor**: monitored mailbox → document classifier → Textract / multimodal read → structured extraction to a typed surety schema (principal, obligee, bond type/amount, per-project WIP rows, working capital) → validation + confidence → a bonding-capacity appetite check → a ranked triage queue, fully instrumented with confidence scores and an audit log. It maps one-to-one to how the market leaders describe their own systems, while the surety focus signals what a generic vendor cannot do.

KEY RESOURCES · TO STUDY FIRST

- **AWS ML Blog** - IDP with Textract + Bedrock + LangChain (closest cloud reference). aws.amazon.com/blogs/machine-learning/intelligent-document-processing-with-amazon-textract-amazon-bedrock-and-langchain/
- **Unstract** - open-source IDP to JSON platform, read as a blueprint. github.com/Zipstack/unstract
- **Google Cloud x Cytora** - most detailed public architecture (golden schema, chain-of-thought). cloud.google.com/blog/topics/financial-services/cytora-uses-generative-ai-to-assess-underwriting-risk
- **SortSpoke / ResourcePro** - vendor-neutral domain and terminology grounding. sortspoke.com/blog/underwriting-submission-triage-explained
- **SFAA + IRMI** - surety fundamentals and how underwriters read the WIP. surety.org/surety-fidelity/what-is-surety/ · irmi.com

Method: 5 search angles → 25 sources fetched → 116 candidate claims → 25 load-bearing claims each verified by 3 independent adversarial reviewers (killed on 2 of 3 refutes). 24 confirmed, 1 refuted (the "deterministic extraction" claim, corrected above). Full cited report: [RLI-submission-ingestion-research.md](#).